

Scheduled FOMC Statements and Cryptocurrency Trading Activity: Intraday Evidence from a 24/7 Market

Authors and affiliations:

1. Manlu Yang

(Institute of Chinese Financial Studies, Southwestern University of Finance and Economics, Ph.D candidate. Primary research areas: Industrial Transformation, Financial Risks, Industrial Economics, Corporate Finance, Green Finance, Environmental Economics. E-mail: yymsherry@163.com)

2. Yufeng Wang

(Professor, College of Management, Sichuan Agricultural University, Ph.D in Economics. Primary research areas: Corporate Finance, Rural Finance. Email: wangyf@sicau.edu.cn)

Corresponding author: Manlu Yang, yymsherry@163.com, SWUFE

ORCID: 0009-0003-1384-9150

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Data and code availability: All data used in this study are publicly available. Scheduled FOMC statement dates are obtained from the Federal Reserve Board's meeting calendar. Hourly cryptocurrency OHLCV data are retrieved from Coinbase Exchange's and Bitfinex's public candles endpoints. The federal funds target range upper bound used for decision classification is retrieved from FRED (series DFEDTARU). The full replication code and downloaded raw data files are included in the submission package.

Highlights

- Scheduled FOMC statements trigger sharp intraday crypto market reactions.
- Bitcoin and Ethereum volatility jumps in the first post-statement hour.
- Trading volume more than doubles around 14:00 ET on statement days.
- Placebo tests, fixed effects, and cross-venue replication support robustness.
- Implications for event risk management and liquidity provision in crypto markets.

Keywords: FOMC; monetary policy announcements; cryptocurrency; intraday volatility; trading volume

Scheduled FOMC Statements and Cryptocurrency Trading

Activity: Intraday Evidence from a 24/7 Market

Abstract

We study how scheduled U.S. monetary policy communication affects cryptocurrency markets in a continuous-trading setting. Combining 41 scheduled FOMC meeting statements (January 2021–January 2026) with hourly BTC–USD and ETH–USD data, we document sharp, concentrated jumps in both volatility and trading activity in the first hour after the 14:00 (New York time) statement release. For Bitcoin, mean absolute hourly returns rise from 0.66% in the hour before the announcement to 1.25% in the first post-announcement hour, while USD trading volume increases by a factor of 2.54. Ethereum exhibits similar effects (0.85% to 1.50%; volume factor 2.81). Matched-week controls, hour-matched placebos, and fixed-effects regressions confirm that these jumps are not explained by intraday seasonality or slow-moving time variation. Cross-venue replication on Bitfinex and robustness to alternative volatility and volume proxies reinforce the conclusion that major cryptocurrencies exhibit sizeable and predictable macro event risk at scheduled FOMC communication times, with direct implications for risk management and liquidity provision.

Keywords: FOMC; monetary policy announcements; cryptocurrency; intraday volatility; trading volume

JEL: E52; G12; G14

1. Introduction

Monetary policy communication is one of the most important and recurrent sources of information arrivals in financial markets. A broad literature shows that macroeconomic announcements and central bank releases generate abrupt changes in volatility and trading activity as investors update beliefs and re-optimize positions (Ederington & Lee, 1993). Among these events, the Federal Open Market Committee (FOMC) statement is particularly salient because it is scheduled, closely followed by global investors, and

typically summarises both the policy decision and the Committee's assessment of economic conditions.

Cryptocurrency markets provide a complementary laboratory for studying announcement effects because they trade continuously. In conventional assets, a key empirical challenge is that the announcement window can intersect with market closures or with opening mechanics, complicating intraday alignment and the measurement of pre-versus post-announcement outcomes. In contrast, the hour before and the hour after 14:00 New York time are both active-trading hours in a 24/7 crypto market on the same calendar date. This allows an unusually clean design: we can compare adjacent hourly candles around a known announcement time while holding constant the day, the local time-of-day, and the prevailing continuous-trading environment.

At the same time, the extent to which major cryptocurrencies price macroeconomic information remains debated. Bitcoin and Ethereum are increasingly viewed as macro-linked risky assets, yet the strength and stability of their responses to U.S. monetary policy news may depend on market structure, investor composition, and the monetary policy regime. Prior work documents heterogeneous crypto reactions to monetary policy announcements and emphasises the speculative nature of crypto assets relative to traditional safe-haven narratives (Baur et al., 2018; Corbet et al., 2020). However, evidence on *announcement-time trading activity* in recent samples—especially using a uniform intraday window and a transparent, easily replicated design—remains limited.

This paper provides a concise set of stylised facts that align with the aims of *Finance Research Letters*: short, timely, and reproducible evidence with clear implications for practitioners and researchers. We combine (i) scheduled FOMC statement dates from the Federal Reserve's meeting calendar with (ii) hourly BTC–USD and ETH–USD OHLCV data from Coinbase. Our core design quantifies within-meeting jumps in volatility and trading activity from the hour immediately before the 14:00 ET statement release to the first hour after. Because the statement time is announced well in advance, any predictable footprint we document is about the *timing* of elevated risk and liquidity demand at a scheduled macro event, rather than about forecasting the direction of returns.

Our main finding is economically large and sharply concentrated. For Bitcoin, mean absolute hourly returns increase from 0.66% in the pre-announcement hour to 1.25% in the announcement hour (a 0.59 percentage-point jump), and USD trading volume increases by a factor of 2.54. Ethereum shows similar patterns (0.85% to 1.50%; volume factor 2.81). The event-time profile (Figure 1) reveals that the spike is localised to the announcement hour, consistent with rapid information processing rather than a gradual drift. Distributional evidence (Figure 2) shows that the effects are not driven by a few outliers.

We deliberately keep the baseline estimand simple—adjacent-hour changes around a known event time—to minimise modelling assumptions and limit researcher degrees of freedom. At the same time, we address the most common identification concerns in layered fashion. Matched-week controls compare each meeting’s jump to the same timestamp one week earlier (same weekday and local time), which absorbs persistent intraday patterns and weekly rhythms. Hour-matched placebos compare 14:00 ET jumps on statement days to 14:00 ET jumps on non-FOMC weekdays, directly addressing intraday seasonality. Fixed-effects regressions using the full hour-level panel control for hour-of-day, day-of-week, and year-month effects with HAC standard errors. Finally, we replicate the headline results on Bitfinex to assess cross-venue stability and show that volume spikes are not mechanically driven by the USD price component by repeating the analysis using base-asset volume.

The contribution is practical and twofold. First, we quantify scheduled *macro event risk* in two benchmark crypto assets in a continuous-trading environment, using a clean and replicable empirical design. Second, by focusing on both volatility and trading activity, we provide evidence relevant for liquidity provision, market making, and intraday risk management around a widely watched macro event. In particular, the results indicate that even market participants who do not trade traditional fixed-income instruments face systematic exposure to scheduled FOMC communication times through crypto positions.

2. Data and method

2.1. FOMC statement times and event timestamps

We obtain scheduled FOMC meeting statement dates from the Federal Reserve's meeting calendar. For each statement date, we define the announcement timestamp as 14:00 New York time and convert it to UTC, explicitly accounting for daylight saving time. This conversion is important because the UTC timestamp corresponding to 14:00 ET changes across the year. The resulting sample contains 41 scheduled statements from January 27, 2021 through January 28, 2026.

The “announcement hour” is defined as the one-hour window starting at 14:00 ET. This hour mechanically contains the statement release and the initial market digestion period; it also overlaps with the start of the press conference (typically 14:30 ET). At hourly frequency, small variation in the exact statement release minute within the hour does not change the event definition. We interpret the announcement hour as the main FOMC communication window faced by intraday crypto market participants.

2.2. Crypto prices and volume

We use Coinbase Exchange's public API to download hourly OHLCV candles for BTC–USD and ETH–USD. The API reports volume in the base asset; we convert it to a USD notional proxy using the hourly close price. Our baseline variables are:

- Hourly log return: $(r_t = (P_t) - (P_{t-1}))$ (aligned to the candle's start time).
- Volatility proxy: $(|r_t|)$.
- USD volume: $(V^{\text{USD}}_t = v_t)$.

Hourly aggregation provides a transparent mapping between the 14:00 ET statement time and a single post-announcement observation across meetings, even with daylight saving time shifts. It also reduces microstructure noise relative to very high-frequency measures while preserving the immediate nature of the reaction. The replication code de-duplicates timestamps and drops incomplete observations. In the appendix, we consider an alternative intrahour volatility proxy based on the log high–low range $((H_t/L_t))$ and an alternative volume proxy based on base-asset volume rather than USD notional volume.

2.3. Announcement-hour jump metrics

Let $(t=0)$ denote the first hour after the statement release (the candle starting at 14:00 ET). Let $(t=-1)$ denote the preceding hour. For each meeting (i) , we compute:

- Volatility jump: $(|r_i| = |r_{\{i,0\}}| - |r_{\{i,-1\}}|)$.
- Trading-activity jump: $(V_i = (V^{\{USD\}}_{\{i,0\}} - V^{\{USD\}}_{\{i,-1\}}))$.

For trading activity, we report geometric-mean volume factors (V_i) , which are natural for log ratios and less sensitive to extreme observations than arithmetic ratios. We report meeting-level averages and associated (t) -statistics, and we benchmark the jumps using three complementary designs:

1. **Matched-week control.** Compare the announcement-hour jump to the same local timestamp one week earlier.
2. **Hour-matched placebo.** Compare 14:00 ET jumps on FOMC statement days to 14:00 ET jumps on non-FOMC weekdays.
3. **Fixed-effects regressions.** Estimate the incremental effect of the announcement hour using the full hour-level panel with hour-of-day, day-of-week, and year-month fixed effects and HAC standard errors.

2.4. Implementation and reproducibility

All inputs are public and all computations are scripted. The replication package downloads the Fed calendar, fetches hourly candles from Coinbase and Bitfinex, constructs event timestamps with correct time-zone handling, and reproduces all tables and figures. The goal is to make the evidence easy to verify and extend without proprietary data or manual intervention.

3. Main results

3.1. Volatility and volume spikes at 14:00 ET

Table 1 reports large and statistically significant jumps in both volatility and trading activity at scheduled FOMC statement releases. For Bitcoin, mean absolute hourly returns increase from 0.66% in the hour before the statement to 1.25% in the announcement hour, implying an average jump of 0.59 percentage points $((t=3.77))$. USD trading volume rises by a factor of 2.54 in the announcement hour relative to the preceding hour $((t=12.21))$ for the log-volume ratio). Ethereum displays similar patterns,

with a 0.65 percentage-point volatility jump ($t=3.40$) and a $2.81\times$ volume increase ($t=12.11$).

Table 1. Intraday volatility and volume jumps at scheduled FOMC statements.

Asset	N	Mean $ r $ (hour -1, %)	Mean $ r $ (hour 0, %)	$\Delta r $ (0 - -1, pp)	$t(\Delta r)$	Mean vol ratio (USD, 0/-1)	$t[\ln(\text{vol}$ ratio)]	Matched- week $\Delta r $ (pp)	t	Matched- week vol ratio factor	t
BTC	41	0.66	1.25	0.59	3.77	2.54	12.21	0.44	2.23	2.39	7.90
ETH	41	0.85	1.50	0.65	3.40	2.81	12.11	0.50	2.17	2.77	8.77

Note: Hour 0 is the first hour after the 14:00 ET statement release; hour -1 is the preceding hour. $|r|$ denotes absolute hourly log return. USD volume is base-asset volume multiplied by the hourly close. Matched-week uses the same weekday and local time one week earlier. (t)-statistics are based on meeting-level observations.

These magnitudes are economically meaningful at hourly frequency. For liquidity providers, the combination of higher volatility and higher order flow implies elevated inventory and adverse selection risk during the communication window. For risk managers and portfolio managers, the results imply that crypto exposures embed predictable intraday risk concentration at scheduled FOMC times, which may matter for intraday VaR, margining, and execution planning.

Figure 1 helps interpret the size of the effect relative to typical intraday conditions. It plots average volatility and USD volume by New York hour on non-event weekdays and overlays the pre- and post-announcement means on statement days. The announcement hour stands out sharply in both panels, consistent with a concentrated information-processing response rather than normal diurnal variation.

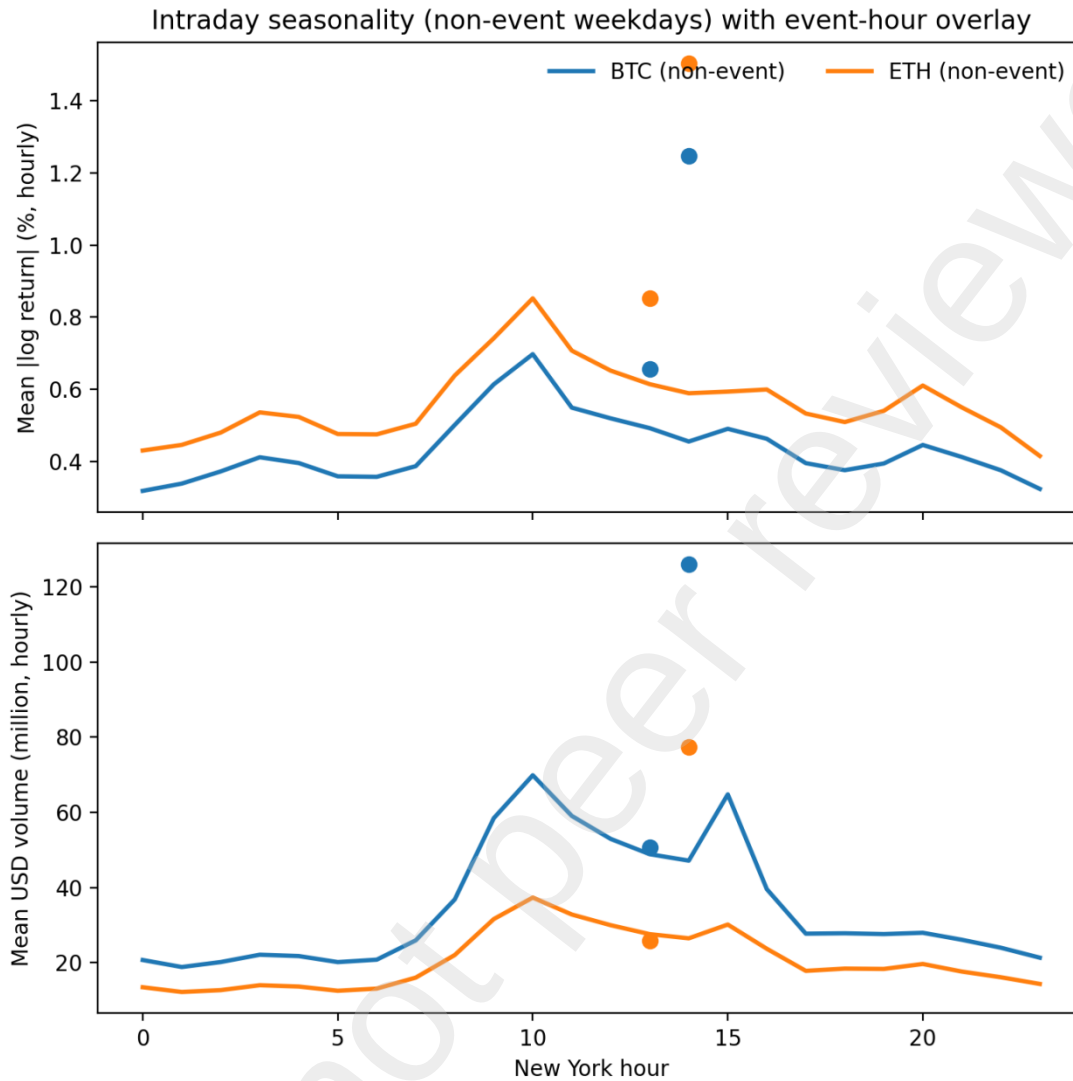


Figure 1. Intraday seasonality and event-hour overlay (New York time).

Note: Lines show average non-event weekday volatility and USD volume by New York hour. Dots overlay the mean values on FOMC statement days for hour -1 (13:00 ET) and hour 0 (14:00 ET).

3.2. Event-time profile and distributional evidence

Figure 2 visualizes the event-time profile from 6 hours before to 6 hours after the statement time. Both assets exhibit a pronounced spike at ($t=0$), consistent with rapid information processing around the scheduled release. In contrast, the hours immediately surrounding the event (e.g., ($t=-2,-1,+1$)) show much smaller average movements, suggesting that the response is not a gradual build-up.

Figure 3 provides distributional evidence by comparing meeting-level announcement-hour jumps to matched-week control jumps. The distributions shift markedly: event-day volatility jumps are higher than control jumps for both BTC and ETH, and the volume-factor distributions are clearly displaced upward. This addresses a common concern that mean effects could be driven by a small number of extreme meetings.

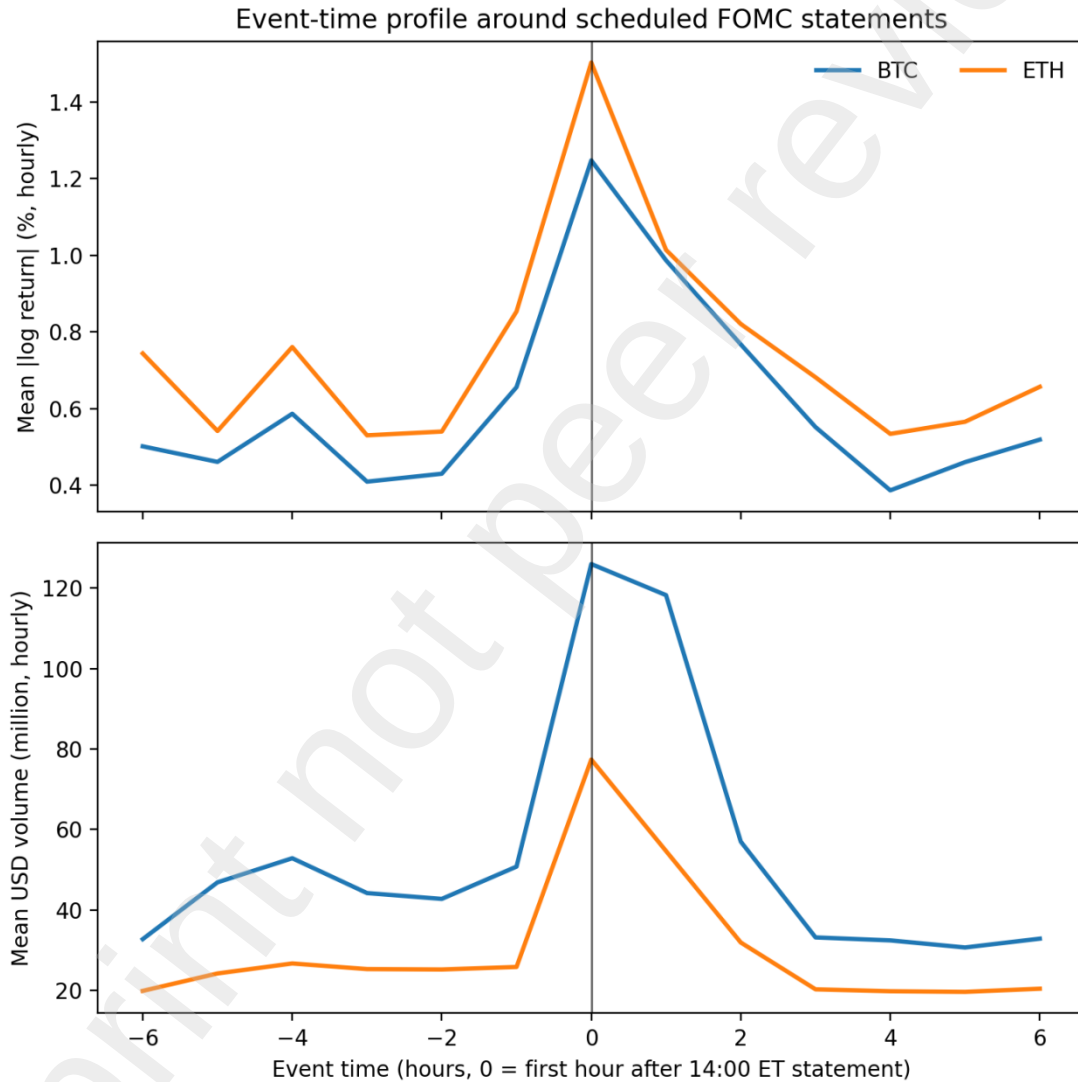


Figure 2. Event-time profile around scheduled FOMC statements.

Note: Mean absolute hourly log returns (top panel) and mean USD trading volume (bottom panel) for BTC–USD and ETH–USD from 6 hours before to 6 hours after the statement release (0 = first hour after 14:00 ET).

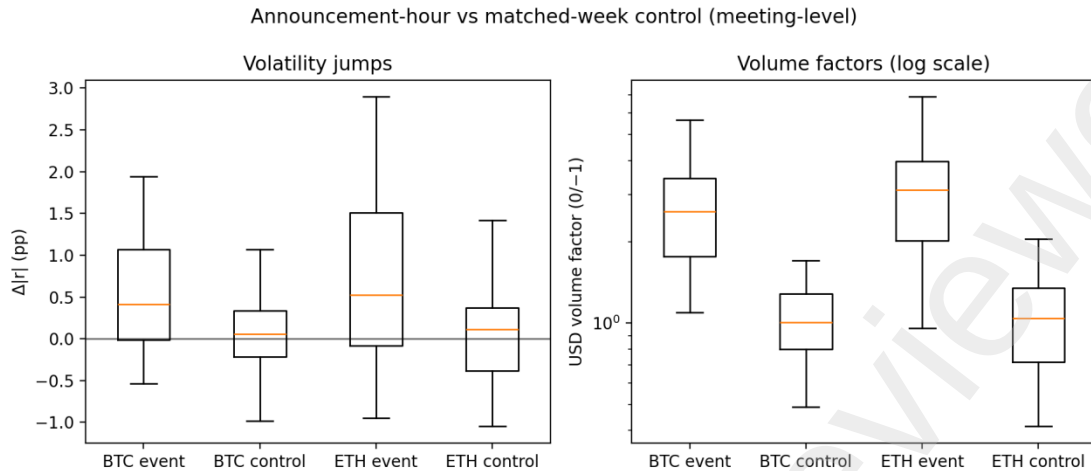


Figure 3. Distribution of announcement-hour jumps vs matched-week control.

Boxplots compare meeting-level volatility jumps ($\Delta|r|$ in pp) and USD volume factors (0/-1) on FOMC statement days to a matched-week control timestamp (same weekday and local time one week earlier).

3.3. Fixed-effects regressions

Table 2 reports hour-level regressions of volatility and volume changes on an indicator for the announcement hour, controlling for hour-of-day, day-of-week, and year-month fixed effects with HAC standard errors. The estimated coefficients closely match the simple within-meeting estimates. For Bitcoin, the regression-implied additional volatility jump at the announcement hour is 0.63 pp ($t=3.99$), and the implied volume factor is 2.62 ($t=12.53$). For Ethereum, the corresponding estimates are 0.67 pp ($t=3.51$) and 2.95 ($t=12.63$). Thus, the evidence for an announcement-time footprint is robust to systematic calendar controls and serial-correlation-robust inference.

Table 2. Regression evidence with fixed effects.

Asset	N (weekday hours)	β on $\Delta r $ (pp)	t	β on $\Delta \ln(\text{USD volume})$	Volume factor	t
BTC	32,739	0.63	3.99	0.96	2.62	12.53
ETH	32,739	0.67	3.51	1.08	2.95	12.63

Note: Hour-level regressions estimate the incremental effect of the announcement hour (14:00 ET on statement days) on the volatility jump ($|r|$) (pp) and the change in log USD volume (V), controlling for hour-of-day, day-of-week, and year-month fixed effects. (t)-statistics use HAC standard errors (24 lags). Volume factor is (V).

4. Robustness and discussion

4.1. Placebo tests and intraday seasonality

Two placebo tests address intraday seasonality concerns. First, the matched-week control (one week earlier at the same local time) shows that announcement-hour jumps remain economically large: volatility jumps exceed the control by 0.44 pp for Bitcoin and 0.50 pp for Ethereum, and volume increases are $2.39\times$ and $2.77\times$ larger than the matched-week benchmark (Table 1). Second, comparing to all non-FOMC weekdays at the same local hour (14:00 ET) yields even stronger rejections of a purely time-of-day explanation (Table A1). The combination of within-hour differencing, matched-week controls, and hour-matched placebos is difficult to reconcile with a pure diurnal-pattern story.

4.2. Cross-venue replication and alternative proxies

To assess whether the results depend on a single exchange's dataset, we replicate the baseline analysis using Bitfinex hourly candles. The announcement-hour footprint remains pronounced for both BTC and ETH (Table A6), indicating that the results are not exchange-specific.

We also consider alternative measurement choices. Table A4 uses the within-hour log high–low range ((H/L)) as an alternative volatility proxy; the announcement-time spike remains sharp and statistically significant. Table A7 repeats the volume-factor analysis using base-asset volume (BTC or ETH units) instead of USD notional volume; the volume spike remains essentially unchanged, indicating that the effect is not mechanically driven by the price component of USD volume.

4.3. Outlier robustness and heterogeneity

Table A5 reports medians and 5% winsorised means for the baseline jump measures. The results remain economically large, suggesting that the conclusions are not driven by extreme outliers.

Scheduled meetings differ in informational content. Some meetings are associated with the Summary of Economic Projections (SEP). Table A3 splits meetings by statement-month (March/June/September/December as a proxy for SEP meetings) and finds sizable jumps in both subsamples. Table A2 provides a descriptive split by the effective policy-rate decision (hike/hold/cut, based on next-day changes in the target

range upper bound). While sample sizes are small for cuts in this period, the announcement-hour footprint is present across decision categories.

4.4. Interpretation and implications

Taken together, the evidence supports the view that major cryptocurrencies are macro-sensitive in the sense that a central, scheduled U.S. monetary policy information event triggers immediate and large changes in both volatility and trading activity. The evidence is intentionally descriptive: we do not attempt to identify policy surprises, separate statement content from subsequent Q&A, or decompose channels (e.g., discount-rate news versus risk-premium news). Instead, we quantify the object most relevant for operational risk and liquidity management: the magnitude and timing of the announcement-hour spike.

For market makers and liquidity providers, the combination of higher volatility and higher flow suggests a period of elevated adverse selection and inventory risk, potentially requiring larger buffers or dynamic quoting policies. For portfolio managers, the results imply that even positions held outside conventional market hours embed predictable event risk at 14:00 ET on scheduled FOMC days. For researchers, the continuous-trading setting highlights the value of crypto markets as a high-frequency laboratory for studying information processing without market-closure confounds.

An immediate practical application is intraday risk forecasting. Many short-horizon volatility and liquidity models for crypto assets are purely time-series based; our evidence suggests that conditioning on scheduled macro communication times can materially shift the conditional distribution of both absolute returns and trading activity. Incorporating a simple indicator for the FOMC statement hour—or a small set of macro-event indicators more broadly—can therefore improve intraday risk limits, margining policies, and execution constraints. The results may also inform option-implied volatility dynamics and the design of event-driven hedging strategies in crypto derivatives markets, where predictable spikes in realised volatility around a known announcement time can affect pricing and risk management.

Two limitations are worth noting. First, our data are hourly candles rather than order-book measures such as spreads or depth. Second, hourly frequency combines the immediate statement reaction with any additional information released during the

subsequent minutes, including the press conference start. Finer sampling could separate these components but would require additional microstructure adjustments and data sources. These limitations do not undermine the central stylised fact: a large and concentrated volatility-and-volume footprint at a widely scheduled macro communication time.

5. Conclusion

Using a clean 24/7 trading setting, we show that scheduled FOMC statements generate sharp and concentrated spikes in cryptocurrency volatility and trading volume. The first post-statement hour exhibits roughly double the absolute return and more than double the trading volume relative to the preceding hour for both Bitcoin and Ethereum. These patterns remain under matched-week controls, hour-matched placebos, and fixed-effects regressions, and they are robust to alternative volatility proxies, outlier treatments, base-volume measures, and cross-venue replication. Overall, the evidence indicates that major cryptocurrencies embed predictable, sizeable macro event risk at scheduled FOMC communication times—an important consideration for risk management, liquidity provision, and empirical modelling of crypto markets. Given leverage and margining in many crypto venues, such scheduled spikes can be operationally consequential even for passively held exposures.

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Appendix (Online)

A1. Hour-matched placebo (intraday seasonality)

This appendix compares announcement-hour jumps to the same local hour (14:00 New York time) on *non-FOMC weekdays* within the sample. For each hour (t) with local time 14:00, we compute:

- $(|r_t| = |r_t| - |r_{t-1}|)$
- $(V_t = (V^{\{USD\}}_t) - (V^{\{USD\}}_{t-1}))$

where (t) is the candle starting at 14:00 local time and (t-1) is the preceding hour. We then compare these jumps on scheduled FOMC statement days to non-FOMC weekdays at the same local hour. Differences use Welch unequal-variance (t)-tests.

Table A1. Hour-matched placebo: 14:00 ET on FOMC vs non-FOMC weekdays.

Asset	N event hours	N non-event hours	Mean $\Delta r $ (event, pp)	Mean $\Delta r $ (non-event, pp)	Diff (event – non-event, pp)	t	p	Mean vol ratio (event)	Mean vol ratio (non-event)
BTC	41	1323	0.59	-0.03	0.62	3.95	0.00030	2.54	0.97
ETH	41	1323	0.65	-0.02	0.67	3.47	0.00123	2.81	0.95

Note: $\Delta|r|$ is in percentage points (pp). “Vol ratio” is $(V^{\{USD\}}_t / V^{\{USD\}}_{t-1})$.

A2. Split by the effective policy-rate change (hike/hold/cut)

We classify meetings using the change in the upper bound of the federal funds target range (FRED series DFEDTARU) from the statement date to the next calendar day in the series (effective date). Table A2 reports descriptive averages of announcement-hour jumps by decision category; sample sizes are small, so inference is not emphasized.

Table A2. Announcement-hour jumps by decision category.

Asset	Decision	N	Mean $\Delta r $ (pp)	Mean vol ratio
BTC	hike	11	0.73	3.00
BTC	hold	24	0.55	2.13
BTC	cut	6	0.53	3.81
ETH	hike	11	0.94	3.73
ETH	hold	24	0.52	2.30
ETH	cut	6	0.63	3.67

A3. Split by SEP vs non-SEP meetings

Some scheduled meetings are associated with the Summary of Economic Projections (SEP). We proxy SEP meetings by statement-month $\in \{\text{March, June, September, December}\}$. Table A3 shows that the announcement-hour footprint remains present in both subsamples.

Table A3. Announcement-hour jumps by SEP vs non-SEP meetings.

Asset	Group	N	Mean $\Delta r $ (pp)	Mean vol ratio (USD, 0/–1)
BTC	SEP meetings (Mar/Jun/Sep/Dec)	20	0.66	2.78
BTC	Non-SEP meetings	21	0.53	2.33

Asset	Group	N	Mean $\Delta r $ (pp)	Mean vol ratio (USD, 0/-1)
ETH	SEP meetings (Mar/Jun/Sep/Dec)	20	0.58	3.08
ETH	Non-SEP meetings	21	0.71	2.57

A4. Alternative volatility proxy: within-hour high–low range

We compute within-hour log range volatility as $((H_t/L_t))$ and repeat the announcement-hour jump analysis. Table A4 indicates that the spike is also visible in within-hour ranges and remains significant relative to the matched-week control.

Table A4. High–low range volatility around scheduled statements.

Asset	N	Mean $\ln(H/L)$ (hour -1)	Mean $\ln(H/L)$ (hour 0)	$\Delta \ln(H/L)$ (0 - -1)	t	Matched-week $\Delta \ln(H/L)$	t
BTC	41	0.0136	0.0275	0.0139	7.00	0.0127	6.25
ETH	41	0.0173	0.0334	0.0160	5.94	0.0148	4.97

A5. Robustness to outliers: medians and winsorised means

Table A5 reports medians and 5% winsorised means for the baseline jump measures. The results remain economically large, indicating that the main conclusions are not driven by a small number of extreme meetings.

Table A5. Robust statistics for announcement-hour jumps.

Asset	N	Median $\Delta r $ (pp)	Median vol ratio (USD, 0/-1)	Winsor(5%) $\Delta r $ (pp)	Winsor(5%) mean vol ratio (USD, 0/-1)
BTC	41	0.41	2.59	0.56	2.54
ETH	41	0.53	3.12	0.66	2.82

A6. Cross-venue replication on Bitfinex

To assess whether the results depend on a single exchange's data, we replicate the baseline analysis using Bitfinex hourly candles (BTC–USD and ETH–USD). Table A6 reports the same announcement-hour statistics as Table 1. The volatility and volume jumps remain large and statistically significant.

Table A6. Bitfinex replication (hourly candles).

Asset	N	Mean $ r $ (hour -1, %)	Mean $ r $ (hour 0, %)	$\Delta r $ (pp)	t	Mean vol ratio (USD, 0/-1)	t	Matched-week $\Delta r $ (pp)	t	Matched-week vol factor	t
BTC	41	0.65	1.24	0.59	3.82	3.36	7.19	0.44	2.30	2.91	4.61
ETH	41	0.83	1.51	0.68	3.58	3.73	7.90	0.56	2.40	3.58	4.79

A7. Base-asset volume (units) instead of USD notional volume

USD notional volume multiplies base-asset volume by the contemporaneous USD price. To confirm that the volume spike is not mechanically driven by the price component, we repeat the announcement-hour analysis using *base-asset* volume (BTC or ETH units). Table A7 shows that the announcement-hour volume factor is essentially unchanged.

Table A7. Announcement-hour volume factors using base-asset volume.

Asset	N	Mean base vol ratio factor (0/-1)	t	Matched-week base vol factor	t
BTC	41	2.53	12.12	2.38	7.82
ETH	41	2.79	11.97	2.75	8.71

Note: Volume factors are geometric means ($()$). “Matched-week” is the ratio-of-ratios relative to the same timestamp one week earlier.

A8. Visual cross-venue replication

Table A6 reports the Bitfinex replication in tabular form. Figure A1 provides a meeting-level visual comparison of announcement-hour volatility and volume changes across venues. Points lie close to the 45-degree line, indicating that the announcement-hour footprint is not driven by exchange-specific measurement.

Cross-venue replication (meeting-level): Coinbase vs Bitfinex

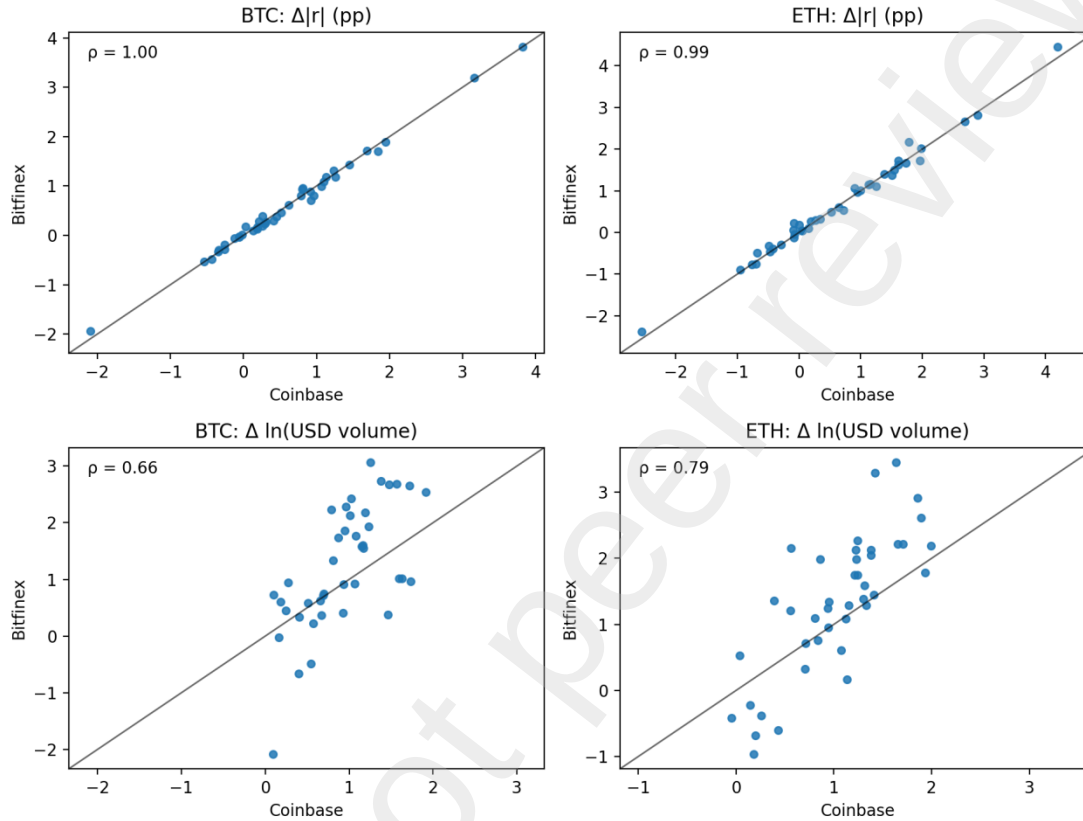


Figure A1. Cross-venue replication (Coinbase vs Bitfinex).

Scatter plots compare meeting-level announcement-hour volatility jumps ($\Delta|r|$, pp) and changes in log USD volume ($\Delta \ln V$) computed from Coinbase candles to the same quantities computed from Bitfinex candles.

Macro event risk in crypto markets: intraday response to scheduled FOMC statements

41 meetings (Jan 2021 to Jan 2026) | BTC-USD and ETH-USD hourly data | 24/7 continuous trading

Event time: 14:00 ET Window: 0h versus -1h Cross-venue replication: Bitfinex

Research design

- Event: scheduled FOMC statement release at 14:00 New York time.
- Sample: 41 scheduled statements from January 2021 to January 2026.
- Market: BTC-USD and ETH-USD hourly OHLCV from Coinbase.
- Core estimand: announcement hour (14:00 to 15:00) minus pre-hour (13:00 to 14:00).

Event-window design

Continuous 24/7 trading allows clean before-versus-after comparison around the same scheduled timestamp.
FOMC statement @ 14:00 ET



Robustness and relevance

Identification
Matched-week controls and hour-matched placebo tests isolate the announcement-time effect from regular/intraday seasonality.

Stability
Fixed effects with HAC standard errors confirm strong significance; cross-venue replication on Bitfinex yields similar patterns.

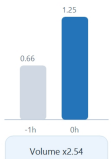
Source: Federal Reserve calendar, Coinbase and Bitfinex public candles, FRED DFEDTARU. Fully reproducible package included.

Main empirical effects

Large and immediate spikes in both volatility and trading activity in the first post-statement hour.

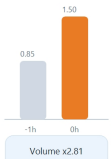
Bitcoin (BTC)

Announcement hour (0h) versus prior hour (-1h)
Mean absolute return
0.66% -> 1.25%
Delta: +0.59 pp | t = 3.77
USD trading volume factor
x2.54
 $\frac{\ln(\text{volume ratio})}{\ln(2.54)} = 12.21$



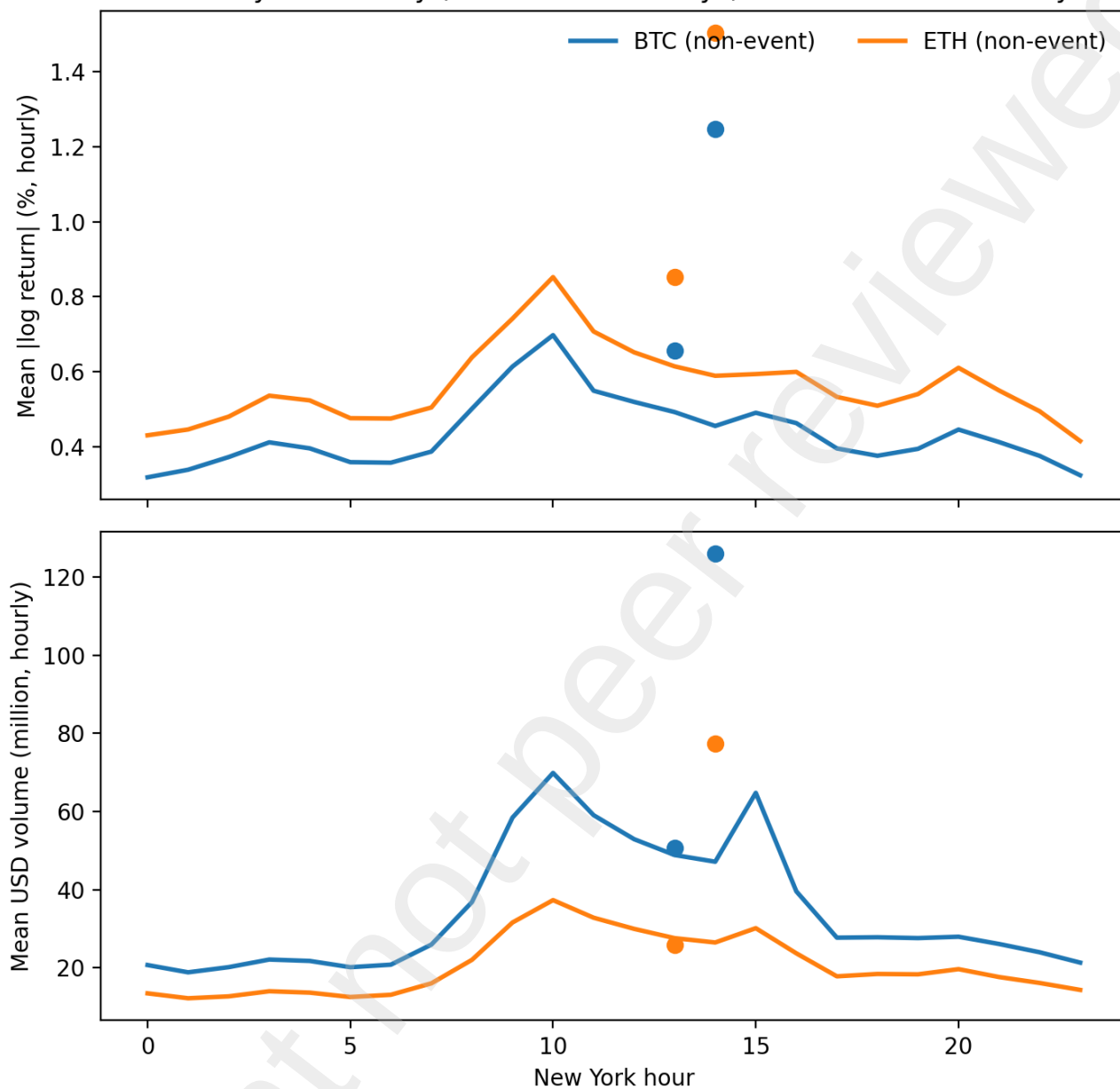
Ethereum (ETH)

Announcement hour (0h) versus prior hour (-1h)
Mean absolute return
0.85% -> 1.50%
Delta: +0.65 pp | t = 3.40
USD trading volume factor
x2.81
 $\frac{\ln(\text{volume ratio})}{\ln(2.81)} = 12.11$

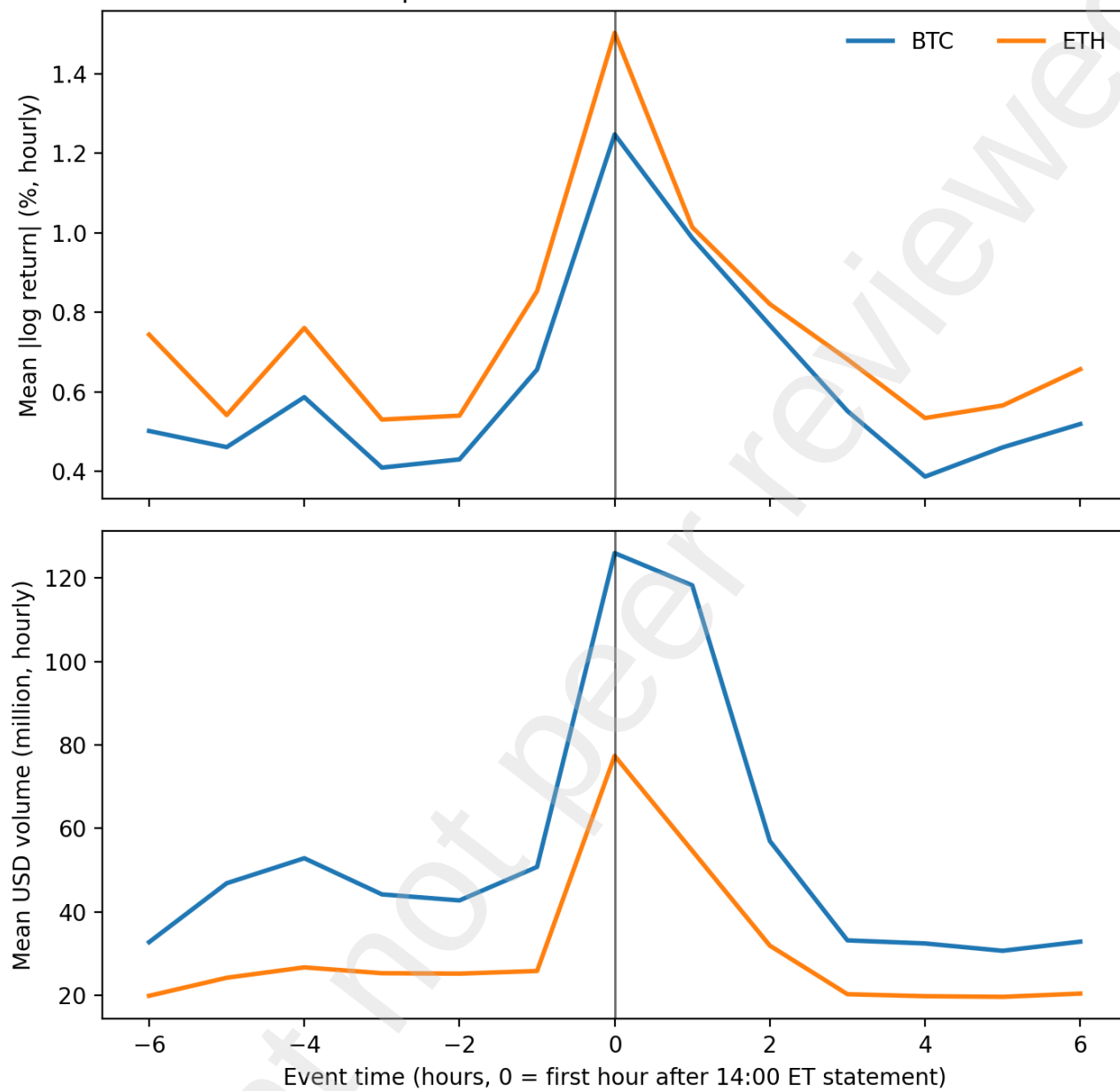


Practical implication
Scheduled FOMC communication windows carry predictable intraday risk and liquidity demand for major crypto assets.

Intraday seasonality (non-event weekdays) with event-hour overlay

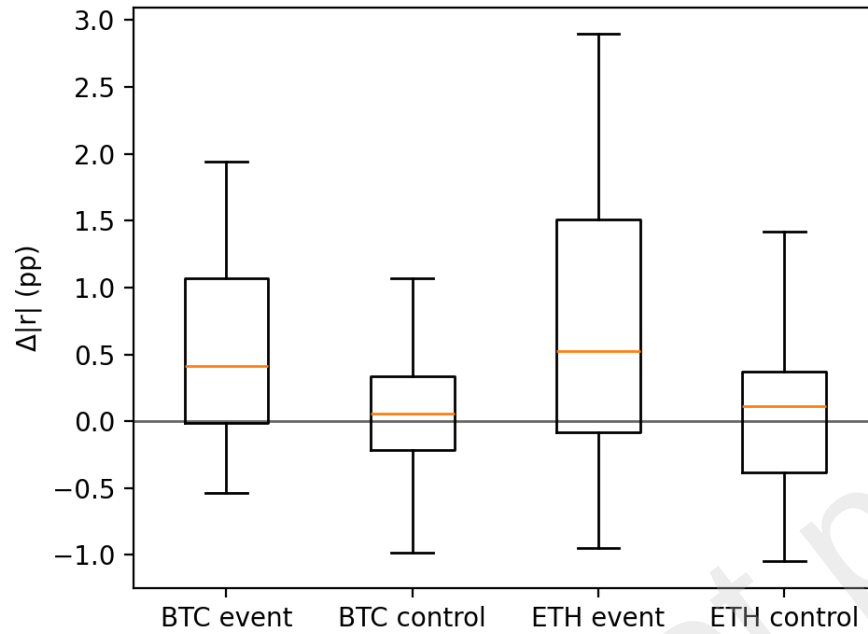


Event-time profile around scheduled FOMC statements

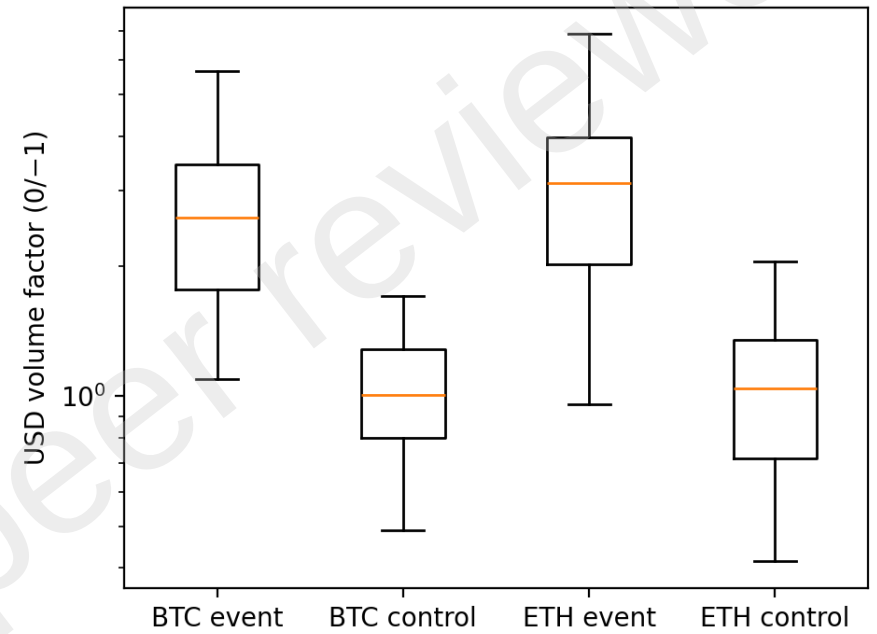


Announcement-hour vs matched-week control (meeting-level)

Volatility jumps

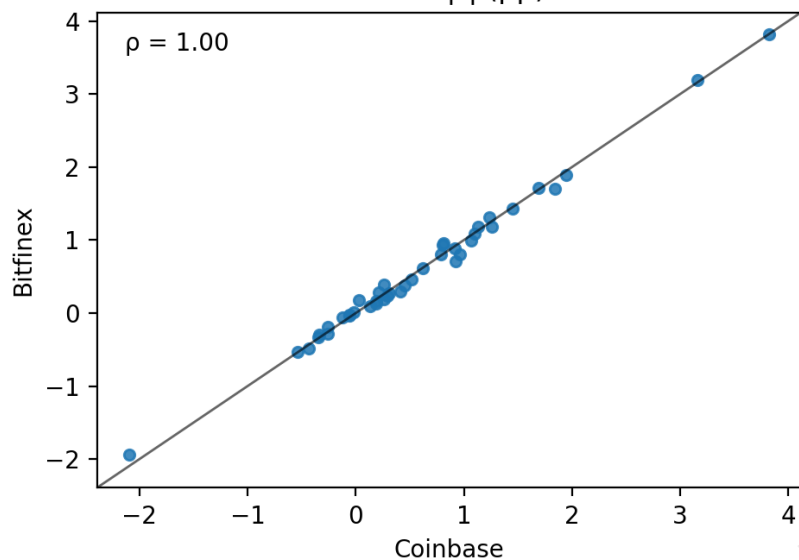


Volume factors (log scale)

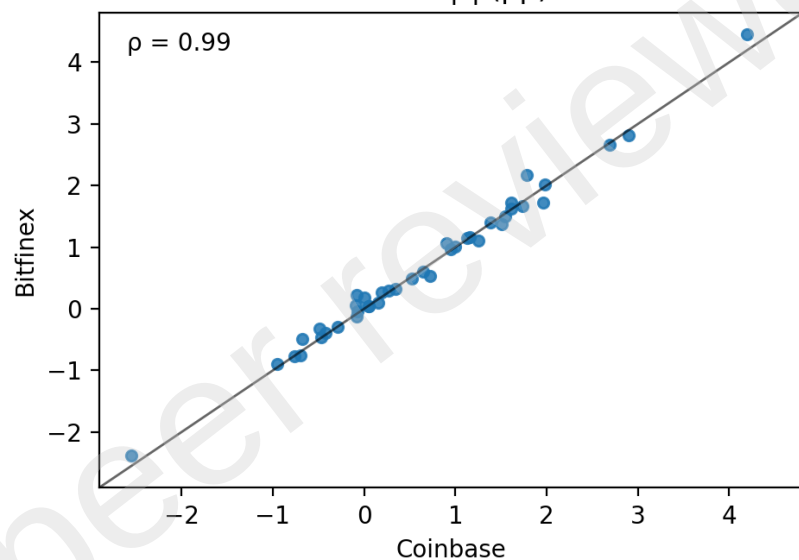


Cross-venue replication (meeting-level): Coinbase vs Bitfinex

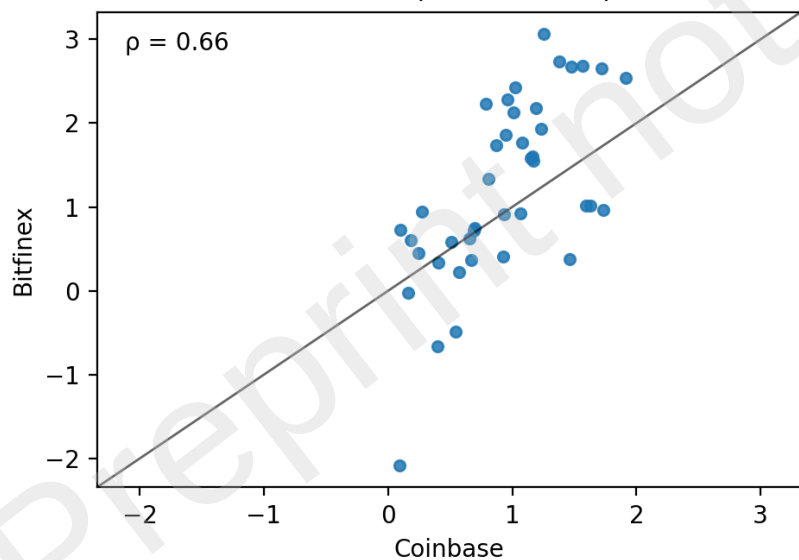
BTC: $\Delta|r|$ (pp)



ETH: $\Delta|r|$ (pp)



BTC: $\Delta \ln(\text{USD volume})$



ETH: $\Delta \ln(\text{USD volume})$

